METRIC

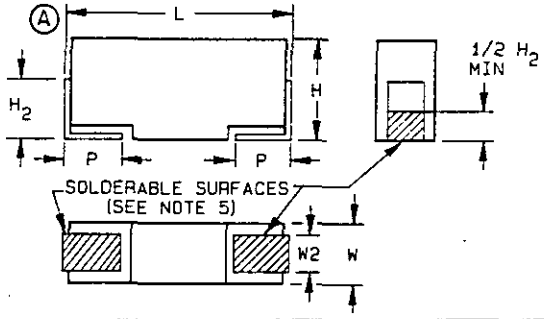
MIL-C-55365/8A
23 August 1990
SUPERSEDING
DOD-C-55365/8
10 March 1987

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLE CWR11 (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.



Dimensions						
Case Code	H (max)	H_2 (min)	L	P ±.3 (.012)	W	W_2 ±.1 (.004) (see note 6)
A	1.8 (.071)	0.7 (.028)	3.2 ±.2 (.126 ±.008)	0.8 (.031)	1.6 ±.2 (.063 ±.008)	1.2 (.047)
B	2.1 (.083)	0.7 (.028)	3.5 ±.2 (.138 ±.008)	0.8 (.031)	2.8 ±.2 (.110 ±.008)	2.2 (.087)
C	2.8 (.110)	1.0 (.039)	6.0 ±.3 (.236 ±.012)	1.3 (.051)	3.2 ±.3 (.126 ±.012)	2.2 (.087)
D	3.1 (.122)	1.0 (.039)	7.3 ±.3 (.287 ±.012)	1.3 (.051)	4.3 ±.3 (.169 ±.012)	2.4 (.094)

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses and are given for information only.
3. These capacitors are designed for mounting by dip or wave soldering, reflow soldering, thermo-compression bonding or by other conventional means.
4. The anode (+) terminal shall be indicated by marking on the case or by bevel on anode end.
A 5. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
A 6. Applicable to solderable surfaces only. The configuration of the nonsolderable area is optional.

FIGURE 1. Style CWR11 (molded) capacitors.

A Denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Sintered tantalum.

Leads: Tab.

Ⓐ Termination finish: In accordance with MIL-C-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

Ⓐ Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Ⓐ Equivalent series resistance (ESR) @ 100 kHz: In accordance with MIL-C-55365. See table I.

Resistance to soldering heat: In accordance with MIL-C-55365.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	- See table I.
	Cap.	- Within tolerance of table I value.
	DF	- See table I.
	ESR	- See table I.
Step 2 (-55°C):	ΔCap.	- Shall not exceed ±10 percent of initial measurement.
	DF	- See table I.
Step 3 (+25°C):	DCL	- See table I.
	ΔCap.	- Shall not exceed ±5 percent of initial +25°C measurement.
	DF	- See table I.
Step 4 (+85°C):	DCL	- See table I.
	ΔCap.	- Shall not exceed ±10 percent of initial +25°C measurement.
	DF	- See table I.
Step 5 (+125°C):	DCL	- See table I.
	ΔCap.	- Shall not exceed ±12 percent of initial +25°C measurement.
	DF	- See table I.

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Step 6 (+25°C): DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial +25°C measurement.
DF - See table I.
ESR - See table I.

Life: Method 108 of MIL-STD-202.

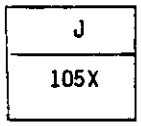
2,000 hours:

At +25°C: DCL - See table I.
ΔCap. - Within ±5 percent of initial measured value.
DF - See table I.
ESR - See table I.

At +85°C: DCL - See table I.

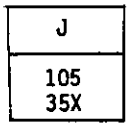
At +125°C: DCL - See table I.

Ⓐ Marking: In accordance with MIL-C-55365. Minimum marking shall be marked on top of case as follows. Print orientation is optional.



(+) polarity stripe, J = JAN
- capacitance in picofarad code; manufacturer's identifier

Case code A



(+) polarity stripe, J = JAN
- capacitance in picofarad code
- rated voltage; manufacturer's identifier

Case codes B, C, and D

TABLE I. Style CWR11 characteristics.

Ⓐ PIN 1/	DC rated voltage (+85°C)	Cap.	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case code
		(nom)	+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11D-225--	6	2.2	0.5	5.0	6.0	6	6	9	15.0	A
CWR11D-685--	6	6.8	0.5	5.0	6.0	6	6	9	4.5	B
CWR11D-156--	6	15.0	0.9	9.0	10.8	6	6	9	3.0	C
CWR11D-476--	6	47.0	2.8	28.0	33.6	6	6	9	1.8	D
CWR11F-155--	10	1.5	0.5	5.0	6.0	6	6	9	15.0	A
CWR11F-475--	10	4.7	0.5	5.0	6.0	6	6	9	4.5	B
CWR11F-156--	10	15.0	1.5	15.0	18.0	6	6	9	2.5	C
CWR11F-336--	10	33.0	3.3	33.0	39.6	6	6	9	1.8	D
CWR11H-105--	15	1.0	0.5	5.0	6.0	4	6	6	15.0	A
CWR11H-335--	15	3.3	0.5	5.0	6.0	6	8	9	5.0	B
CWR11H-106--	15	10.0	1.6	16.0	19.2	6	8	9	2.5	C
CWR11H-226--	15	22.0	3.3	33.0	39.6	6	8	9	1.8	D

See footnote at end of table.

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TABLE I. Style CWR11 characteristics - Continued.

PIN 1/	DC rated voltage (+85°C)	Cap.	DC leakage (max)			Dissipation factor (max)			Max ESR 100 kHz	Case code
		(nom)	+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	+25°C	
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR11J-684--	20	0.68	0.5	5.0	6.0	4	6	6	15.0	A
CWR11J-225--	20	2.2	0.5	5.0	6.0	6	8	9	6.0	B
CWR11J-475--	20	4.7	1.0	10.0	12.0	6	8	9	3.0	C
CWR11J-156--	20	15.0	3.0	30.0	36.0	6	8	9	1.8	D
CWR11K-474--	25	0.47	0.5	5.0	6.0	4	6	6	18.0	A
CWR11K-155--	25	1.5	0.5	5.0	6.0	6	8	9	7.0	B
CWR11K-335--	25	3.3	0.9	9.0	10.8	6	8	9	4.0	C
CWR11K-106--	25	10.0	2.5	25.0	30.0	6	8	9	2.0	D
CWR11M-104--	35	0.10	0.5	5.0	6.0	4	6	6	24.0	A
CWR11M-154--	35	0.15	0.5	5.0	6.0	4	6	6	21.0	A
CWR11M-224--	35	0.22	0.5	5.0	6.0	4	6	6	19.0	A
CWR11M-334--	35	0.33	0.5	5.0	6.0	4	6	6	18.0	A
CWR11M-474--	35	0.47	0.5	5.0	6.0	4	6	6	11.0	B
CWR11M-684--	35	0.68	0.5	5.0	6.0	4	6	6	8.0	B
CWR11M-105--	35	1.0	0.5	5.0	6.0	4	6	6	7.0	B
CWR11M-155--	35	1.5	0.5	5.0	6.0	6	8	9	6.0	C
CWR11M-225--	35	2.2	0.8	8.0	9.6	6	8	9	4.0	C
CWR11M-335--	35	3.3	1.2	12.0	14.4	6	8	9	3.0	D
CWR11M-475--	35	4.7	1.7	17.0	20.4	6	8	9	2.5	D
CWR11N-104--	50	0.10	0.5	5.0	6.0	4	6	6	22.0	A
CWR11N-154--	50	0.15	0.5	5.0	6.0	4	6	6	19.0	B
CWR11N-224--	50	0.22	0.5	5.0	6.0	4	6	6	16.0	B
CWR11N-334--	50	0.33	0.5	5.0	6.0	4	6	6	13.0	B
CWR11N-474--	50	0.47	0.5	5.0	6.0	4	6	6	9.0	C
CWR11N-684--	50	0.68	0.5	5.0	6.0	4	6	6	7.0	C
CWR11N-105--	50	1.0	0.5	5.0	6.0	4	6	6	6.0	C
CWR11N-155--	50	1.5	0.8	8.0	9.6	6	8	9	5.0	D
CWR11N-225--	50	2.2	1.1	11.0	13.2	6	8	9	3.5	D

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, and failure rate level.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 17, 99
NASA - NA
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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